

Image classification

- **Data clustering** is a mathematical process of classification of any type of data into classes; the data in each class are equal or close together in their value.
- **Image classification (spectral clustering)** is classifying pixels of a multispectral image into discrete classes based on the value of pixel reflectance or digital number (DN).
- Use **Unsupervised** classification to cluster pixels in a dataset (image) based on **statistics only**, without any user-defined training classes. Two unsupervised classification techniques are available:
 - 1- ISODATA Classification.
 - 2- K-Means Classification.
- Use **Supervised** classification to cluster pixels in a dataset (image) into classes based on user-defined **training classes (region of interest, RIO)**. Following are some popular supervised classification methods available in ENVI:
 - 1- Parallelepiped Classification.
 - 2- Minimum Distance Classification.
 - 3- Mahalanobis Distance Classification.
 - 4- Maximum Likelihood Classification.
 - 5- Binary Encoding Classification.
 - 6- Spectral Angle Mapper Classification.
 - 7- Spectral Information Divergence Classification.
- Some of these supervised classification methods (no.1, 2, 3, 4, 5) are based on the **training classes (region of interest, RIO)**. The difference between them is the algorithm used in the clustering calculation. Methods (6, 7) are based on what is known as “**spectral library**”. These libraries are available in ENVI and represent the reflectance of minerals, rocks or vegetation types that can be used for minerals, rocks, or vegetation detection and classification.

Unsupervised Classification

A- Using ISODATA Classification

1. From the ENVI main menu bar, open the original image to be classified.
2. From the ENVI main menu bar, select **Classification** → **Unsupervised** → **IsoData**. The Input File dialog appears.
3. Select an input file (the image to be classified), then click **OK**. The ISODATA Parameters dialog appears.
4. Enter the minimum and maximum **Number of Classes** you want. ENVI uses a range for the number of classes because the ISODATA algorithm splits and merges classes based on input thresholds and does not keep a fixed number of classes.

ISODATA Parameters

Number of Classes: Min 5 Max 10

Maximum Iterations 1

Change Threshold % (0-100) 5.00

Maximum Stdev From Mean

Maximum Distance Error

Output Result to ☒ File ☐ Memory

Enter Output Filename Choose

Minimum # Pixel in Class 1

Maximum Class Stdev 1.000

Minimum Class Distance 5.000

Maximum # Merge Pairs 2

OK Queue Cancel Help

5. Enter the maximum number of iterations in the **Maximum Iterations** field and a change threshold (0-100%) in the **Change Threshold %** field.

Iteration is the repetition of the classification process to improve the mean value of each class and class limits. **Threshold** is the number of pixels (in percent) below which the classification process stops. It means that the clustering process ends when the number of pixels in each class changes by a percent less than the threshold value. ENVI uses the change threshold to end the iterative process when the number of pixels in each class changes by less than the threshold. The classification ends when either this threshold is met or the maximum number of iterations is reached.

6. Enter the minimum number of pixels needed to form a class in the **Minimum # Pixels in Class** field. If there are fewer than the minimum number of pixels in a class then ENVI deletes that class and the pixels placed in the class(es) nearest to them.
7. Enter the maximum class standard deviation (in DN) in the **Maximum Class Stdev** field. If the standard deviation of a class is larger than the defined standard deviation then the class is split into two classes.

8. Enter the **minimum class distance** (in DN) between class means and the **maximum number of merge pairs** in the fields provided.

If the distance between class means is less than the minimum value entered, then ENVI merges the classes. The maximum number of class pairs to merge is set by the maximum number of merge pairs parameter.

To set the optional standard deviation to use around the class mean and/or the maximum allowable distance error (in DN), enter the values in the **Maximum Stdev From Mean** or **Maximum Distance Error** fields, respectively.

If you enter values for both of these optional parameters, the classification uses the smaller of the two to determine which pixels to classify. If you do not enter a value for either parameter, then all pixels are classified.

9. Select output to **File** or **Memory**.

10. Click **OK**. The status bar cycles from 0 to 100% for each iteration of the classifier. ENVI adds the resulting output to the **Available Bands List**.

11. To display the resulted classification load the classification image in a new display using the “grey scale” option.

B- Using K-Means Classification

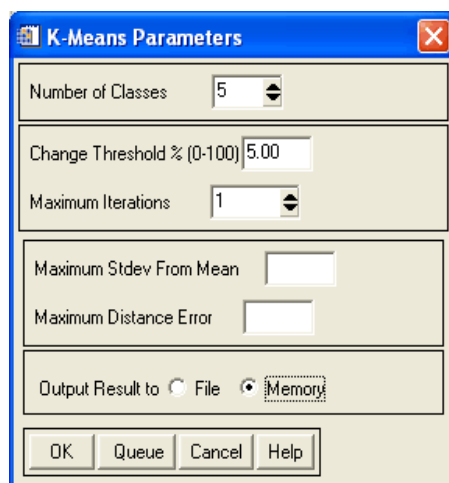
Practically, the steps used to carry out the K-mean classification in ENVI are similar to the isodata classification. The only difference is that the number classes is given a fixed value determined by the user. The number of classes in the isodata classification is a range between a minimum and maximum values determined by the user.

In both types of unsupervised classification methods, do the following:

1- use the maximum iteration = 1
(the default value), and get the Classified image.

2- use the maximum iteration = 5
and get the classified image.

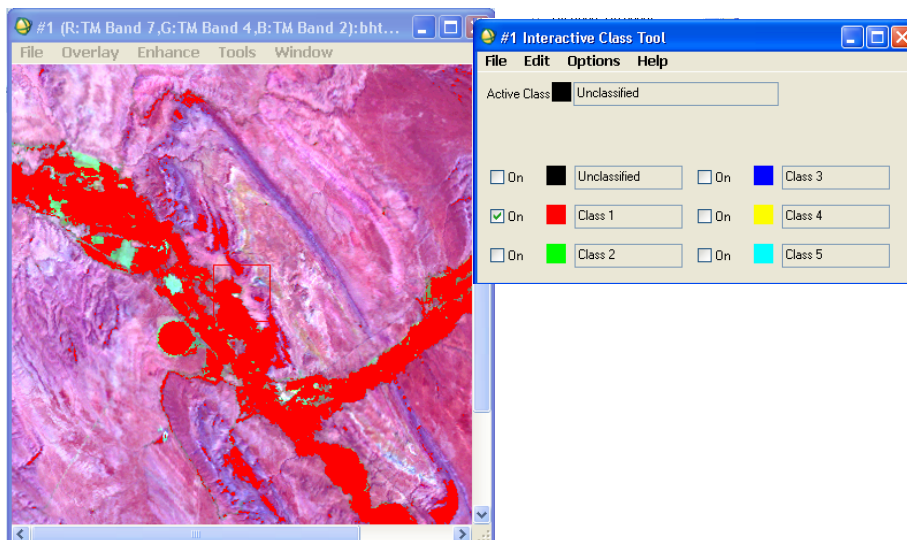
3- Notice the difference between the Two cases.



Overlaying the classified classes on the original image, Edit class names and colors, and Merging classified classes:

You can to interactively overlay classes on and off on the displayed image:

1. From the Available Bands List, load the **original image** as a gray scale image or multispectral image.
2. From the Display group menu bar, select **Overlay → Classification**. The Interactive Class Tool Input File dialog appears.
3. Select the **classified image** file within the “Select Input File section” of this dialog and click **OK**. The Interactive Class Tool appears with each class listed with its corresponding colors.

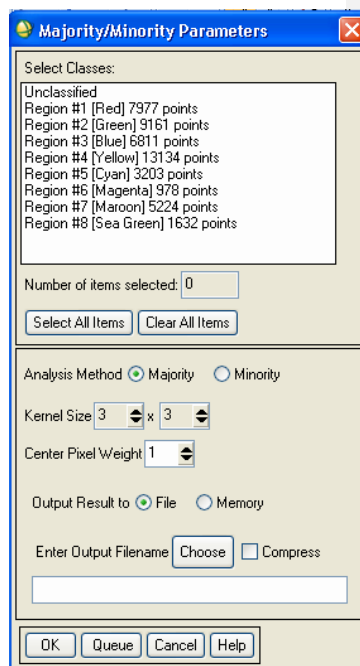


4. Click each **On** check box to change the display of each class as an overlay on the original image.
- 5- To edit the class name or color use Options → **Edit class colors / names**.
- 6- To merge two classes or more use Options → **Merge classes**.
7. From the Display group menu bar, select **File → Save Image As → Image File** to burn in the classes and output to a new file.
8. From the Interactive Class Tool menu bar, select **File → Cancel** to exit the interactive tool.

Improving and smoothing the classified image (solve the problem of isolated pixels occurring in other larger classes.):

There are several methods to solve this problems. Some methods consider the isolated pixels as unclassified and others add them to the surrounding class. The most popular method is the “majority/minority analysis” technique.

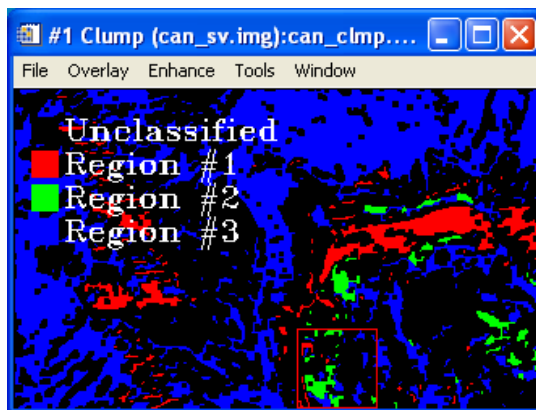
1. From the ENVI main menu bar, select **Classification** → **Post Classification** → **Majority/Minority Analysis**. The Input File dialog appears.
2. Select the classification image file, then click **OK**. The Majority/Minority Parameters dialog appears.
3. In the list of classes, select the classes that you want to apply the analysis to. If the problem of isolated pixels is found in all classes, choose all classes.
4. Select the analysis method, by clicking the corresponding toggle button. Choose “majority”.
5. Enter a kernel size. Larger kernel sizes produce more smoothing of the classification image.
6. Select output to **File** or **Memory**.
7. Click **OK**. ENVI adds the resulting output to the **Available Bands List**.
8. Notice if the smoothing process is ok. If not, repeat the process with a higher kernel value.



Adding Classification Keys (legend) Using Annotation

ENVI provides annotation tools to put classification keys (legend) on images and in map layouts. The classification keys are automatically generated.

1. From the Display group menu bar of the classified image, select **Overlay → Annotation**.
2. From the Annotation menu bar, select **Object → Map Key** to start annotating the image. You can edit the legend characteristics by clicking the **Edit Map Key Items** button in the dialog and changing the desired characteristics.
3. Click once with the left mouse button in the Image window to place the legend in the image window.
4. Click and drag the legend using the left mouse button in the display to place the key.
5. Click in the display with the right mouse button to finalize the position of the legend.



Query/Retrieve Use Cases

Version 1.1

Use Cases

New Patient

1. Get all images related to Patient

Existing patient

1. Get any new images related to Patient:
 - a. All
 - b. Specific
 - i. Only want the PET that was done last week.
2. I want the current plan for patient A which includes structure set and dose
3. I want to find a plan/images/rt structures/dose that was delivered last week Thursday.
4. I want to see all of Dr. Jones's patient.
5. I want to see all plans for this body site. If I'm treating the pelvis I want would want to know if there were any prior treatments that are pelvis or near the pelvis.
6. *for discussion* My machine went down and I want to know which patient is assigned to this machine.
7. I want to know which plan was approved.
8. I want to know all plans for John that has a dose of 4500 cGy prescribed to the target. What was the final dose and total dose to the organ at risk.
9. I want know all plans with fractionation with fewer than 5 fractions so that I know which plans I am delivering large amount of dose. I like to see the target prescription dose as well. So I know how many fractions for which dose.
10. Does he have any restriction on body.
11. Why is he coming back.

From IHE wiki:

- * Retrieve images, structures, plan from CTSim/Archive for planning
- * Retrieve secondary images from an archive or a modality to assist in planning
- * Retrieve plan and doses from other planning systems for comparison
- * Retrieve treatment records and verification images for offline review
- * Retrieve plan and dose for dose accumulation
- * Retrieve a complete set of patient treatment planning data for clinical trials submission

Assumptions

The baseline behavior of the DICOM Query/Retrieve C-FIND operation requires a hierarchical search method. This means that in order to find a certain Series the SCU must do two prior queries to find the previous levels namely Study and Patient. And three for Composite Object Instance queries.

The standard defines an extended relational query behavior for C-FIND SCU and SCP that enables applications to skip the Study and Patient level queries and directly search for Series. This however requires the query to contain the unique keys (Patient ID and Study Instance UID). Some SCP implementations might support universal matching, which means that the SCU must include those two attributes in the query dataset but might leave the Value empty or set to '*'. This also applies to the Composite Object Instance level.

The assumption is that the intended use case can be achieved using either baseline behavior and relational queries. The latter method will minimize the number queries needed, but bottom line the result will be the same.

Proposed flow

New Patient

1. Get all images related to Patient

Existing patient

1. Get any new images related to Patient:
 - a. ALL
 - b. Specific
 - i. Only want the PET that was done last week.
2. I want the current plan for patient A which includes structure set and dose

Commented [SPB1]: Content Retriever sends C-FIND with match key Modality (0008,0060) = 'CT\MR\PT'

This is tricky. Even if the modality depicts an image modality, the actual data might contain something else. Such as raw data. The only way to know for sure that a given Series is a real image is using the SOP Class UID, which is a Composite Instance level attribute. More defined terms can be added for additional modalities,

No additional requirements from profile needed. Default SCU/SCP behavior enough.

Commented [SPB2]: No additional requirements from profile needed. Default SCU/SCP behavior enough.

Commented [SPB3]: What does current really mean? Is it already available to the operator or is it the latest plan for patient A?
Does this mean 'the plan has structure set and dose' or 'I want structure set, plan and dose'?

Main issue here is that the dose references the plan and not the other way around. The Content Receiver must therefore query for all plans before querying for doses that references those plans.

Option 1—Find all plans using various C-FIND on RTSTRUCT/RTPLAN/RTDOSE:

Subtask 1: Find all patients structure set instances

1. C-FIND-SERIES (all structure set series)
Modality = 'RTSTRUCT'
2. C-FIND-INSTANCE (structure set instances)
SOP Class UID = RT Structure Set Storage
Series Instance UID = previous result

Subtask 2: Find all patients plans

3. C-FIND-SERIES (referenced plan series)
Modality (0008,0060) = 'RTPLAN'
Referenced RT Plan Sequence (300C,0002) = result from subtask 1
4. C-FIND-INSTANCE (plan instances)
Series Instance UID = previous result
SOP Class UID = RT Plan Storage

Subtask 3: Find patients doses

5. C-FIND-SERIES (all dose series)
Modality = 'RTDOSE'
6. C-FIND-INSTANCE (dose instances)
Series Instance UID = previous result
SOP Class UID = RT Dose Storage
Referenced RT Plan Sequence (300C,0002) = patient plans (subtask 2)
Dose Summation Type = x
Dose Type = x

Subtask 4: Process result

7. Order plans by Content Date/Time and return first

Option 2—Use RT Plan Geometry:

Subtask 1: Find all patients non-geometric plans

1. C-FIND-SERIES (plan series)
Modality (0008,0060) = 'RTPLAN'
RT Plan Geometry (300A,000C) = 'PATIENT'
2. C-FIND-INSTANCE (plan instances)
Series Instance UID = previous result

Subtask 2: Find patients doses

3. C-FIND-SERIES (all dose series)
Modality = 'RTDOSE'
4. C-FIND-INSTANCE (dose instances)
Series Instance UID = previous result
Referenced RT Plan Sequence (300C,0002) = patients plans (subtask 2)

Subtask 3: Process result

5. Filter and order plans by Content Date/Time and return first

Option 3—Use C-FIND-REFERENCED

Subtask 1: Find all patients plans

1. C-FIND-SERIES
Modality = 'RTPLAN'
2. C-FIND-INSTANCE
SOP Class UID = RT Plan Storage
Series Instance UID = previous result
3. C-FIND-RELATED
SOP Instance UID = previous result
4. Filter and order plans by Content Date/Time and return first

3. I want to find a plan/images/rt structures/dose that was delivered last week Thursday.
4. I want to see all of Dr. Jones's patient.
5. I want to see all plans for this body site. If I'm treating the pelvis I want would want to know if there were any prior treatments that are pelvis or near the pelvis.

Main issue here is fuzzy matching. Treatment site is not well defined in RT Plan.

Commented [SPB4]: Query for treatment records?

Commented [SPB5]: Is this Referring Physician's Name in General Study?

Option 1—Match by Treatment Sites (300A,000B)

1. C-FIND-SERIES (all plans)

Modality = 'RTPLAN'

2. C-FIND-INSTANCE

Series-Instance-UID = previous result

Treatment Sites (300A,000B) = '*pelvis*'

Option 2—Match by Prescription Target (ROI)

Option 3—Match by Beam Isocenter

Commented [SPB6]: This requires fuzzy matching and needs to be negotiated with SCP. Some SCP implementers might use fuzzy matching by default. Otherwise, case-sensitivity and localization will be a problem.

Commented [SPB7]: Just added for completion. Option 2 would match the prescription (Dose Reference) Option 3 (advanced) would match the beam isocenters against the volume of target structures.

6. *for discussion* My machine went down and I want to know which patient is assigned to this machine.

7. I want to know which plan was approved.

C-FIND-INSTANCE Approval-Status = 'APPROVED'

8. I want to know all plans for John that has a dose of 4500 cGy prescribed to the target. What was the final dose and total dose to the organ at risk.

C-FIND-INSTANCE Target-Prescription-Dose = 4500

9. I want to know all plans with fractionation with fewer than 5 fractions so that I know which plans I am delivering large amount of dose. I like to see the target prescription dose as well. So I know how many fractions for which dose.

C-FIND-INSTANCE Number-Of-Fractions-Planned = '0\1\2\3\4\5'

Commented [SPB8]: Range matching only supported for dates. This could be extended to support number range matching (e.g. 0-5).

10. Does he have any restriction on body.

Commented [SPB9]: Is this HIS information? Like patient flags?

11. Why is he coming back.

Commented [SPB10]: Not sure how to translate into C-FIND. Previous diagnosis?